MANUFACTURING PERFORMANCE & KPI DASHBOARD

Portfolio Case Study | PostgreSQL · SQL · Power Query · Power BI · DAX

Project Overview

An end-to-end business intelligence project exploring manufacturing output, production-plan attainment, quality, downtime, and equipment effectiveness using a synthetic manufacturing dataset.

Dataset scope: 3,000 records from 2020–2024, covering 10 plants, 36 production lines, and 49 work centers.

Business Problem

Manufacturing teams need a clear view of whether production targets are being met and where operational losses may occur. This project consolidates core indicators into an interactive dashboard and supports investigation across plants and dates.

- Compare actual production with planned production.
- Identify plants with notable production gaps or downtime costs.
- Monitor defect rates, monthly output, and average OEE.
- Investigate downtime categories and document data-quality limitations.

Technical Approach

PostgreSQL and SQL

Structured the data into a schema with a production fact table and dimensions for date, plant, line, product, shift, and work center.

Power Query

Combined production records with source fields and product keys, added downtime-category fields for investigation, and checked row counts and source-data consistency.

Power BI and DAX

Developed measures for actual and planned production, production variance, defective units, defect rate, unplanned downtime, downtime cost, and average Overall Equipment Effectiveness (OEE). Built KPI cards, comparative charts, a monthly production trend, and plant/date filters.

Selected Findings

Metric	Result
Actual production	1,448,365 units
Planned production	2,219,673 units
Production variance vs. plan	-34.75%
Defective units	3,944
Defect rate	0.27%

Unplanned downtime	190,980.64 minutes
Calculated downtime cost	$28,340,112.58
Average OEE	65.28%

At plant PLT-004, work center WC-039 recorded 1,851.43 minutes of unplanned downtime. Mechanical downtime was the largest of the six recorded categories at 918 minutes, approximately 55.4% of categorized downtime.

Data Quality and Limitations

For PLT-004 / WC-039, the six downtime-category fields summed to 1,658.20 minutes, leaving an unreconciled difference of 193.23 minutes compared with total unplanned downtime. The discrepancy was present in the source data and was not introduced by the Power Query merge.

The dataset is synthetic, and downtime cost is a calculated project metric. The production gap is a signal for further investigation, not proof of a specific cause.

Recommendations

- Investigate recurring mechanical downtime using maintenance logs and failure records.
- Review electrical downtime as the next-largest recorded category.
- Reconcile the 193.23-minute difference and review downtime-coding rules.
- Investigate the production-plan gap before attributing it to a single cause.
- Monitor production, quality, downtime, and OEE over time.

Outcome

The project demonstrates an end-to-end analytics workflow, from relational data modeling in PostgreSQL through data preparation in Power Query to KPI development and interactive reporting in Power BI. It showcases SQL, data validation, DAX, dashboard development, operational analysis, and transparent communication of findings and limitations.

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